

Neval Babu

Right to Work: UK (No sponsorship required) | Leicester | 07459875933 | Neval271@gmail.com

Portfolio: neval-babu-portfolio.netlify.app

SKILLS

Languages & Frontend: Python, Java, C++, JavaScript, TypeScript, HTML, CSS, React, React Hooks, Vite, Pygame

Backend & Data: Node.js, Express, Spring Boot, Spring MVC, Spring Security, PostgreSQL, MySQL, Redis, Prisma ORM, SQL, REST APIs, WebSockets, RSS, JWT, bcrypt

Testing, Cloud & Tools: Vitest, Integration Testing, Postman, Swagger/OpenAPI, Git, GitHub, GitLab, Docker, Microsoft Azure, Azure App Service, Azure Web PubSub, Netlify, Railway, Android Studio, Shell Scripting, OpenAI Codex

Practices: Agile, Scrum, SDLC, Object-Oriented Programming, Responsive Web Design

EDUCATION

BSc (Hons) Computer Science | University of Leicester | **First Class Honours (76+ %)**

PROJECTS

PitchBrief - Personalised Football News Aggregator | www.pitchbrief.co.uk

- Developed a full-stack personalised football news aggregator with React, TypeScript, Node.js, Express and PostgreSQL, bringing club-specific articles from external RSS feeds into one responsive platform.
- Built Express backend services for authentication, favourite clubs and article bookmarks, securing user accounts and personalised features through JWT-protected routes and bcrypt password hashing.
- Designed relational PostgreSQL data models with Prisma ORM to provide structured storage for users, clubs, articles and bookmarks.
- Performed integration testing in Postman to verify backend requests, responses, authentication and error handling.
- Configured Vitest to unit-test core functionality, helping identify regressions and confirm that individual features behaved as expected.
- Deployed the React frontend to Netlify and the Express backend with PostgreSQL to Railway for separate production hosting.

IBM SkillsBuild Companion App

- Developed a gamified web application for IBM SkillsBuild within an 8-member team to create a more engaging learning experience
- Engineered backend services with Java and Spring Boot, exposing REST APIs to handle business logic and application functionality.
- Configured a MySQL relational database for secure data storage and structured user management
- Led the Agile development process as Scrum Master, coordinating sprint activities and team progress to ensure project milestones were met
- Secured application features with Spring Security to support authentication and controlled access.
- Utilised GitLab for version control and collaboration, supporting efficient teamwork and maintaining code consistency

Portfolio Website | neval-babu-portfolio.netlify.app

- Built a responsive portfolio with React, TypeScript and Vite, presenting my skills and software projects clearly across desktop and mobile devices.
- Created reusable React components and TypeScript data structures to dynamically render project information, technology icons and external links, reducing duplication and speeding up future updates.
- Integrated the Intersection Observer API with React Hooks to track visible sections and automatically synchronise the active navigation state while scrolling.
- Managed version control with Git and GitHub, tracking development changes through commits before deploying the production build to Netlify.

Discover Flavour Web Application

- Built a responsive food-discovery website with HTML, CSS and JavaScript, demonstrating core front-end development without relying on a framework.
- Applied JavaScript DOM manipulation and event handling to deliver interactive client-side functionality across desktop and mobile layouts.

Private Cloud Whiteboard

- Distributed a shared, interactive whiteboard application across multiple Microsoft Azure App Service nodes
- Achieved persistent, low-latency connections by integrating Azure Web PubSub for WebSocket communication and Redis Cache for state synchronisation and rapid data retrieval
- Containerised the application with Docker to simplify horizontal scaling, allowing quick instance deployment during peak demand.

Scrabble Game

- Developed a modular desktop game with Python, Pygame and OOP, separating board, tile and game-state logic into reusable components.
- Integrated a DAWG (CompletionDAWG) structure for fast word validation and prefix searches during computer move generation.
- Engineered computer-opponent logic with itertools permutations and adjacency-based board searches to generate and evaluate valid word placements.
- Applied Python multithreading to run move calculations outside the main Pygame event loop, keeping the interface responsive during processing.

WORK EXPERIENCE

Amazon Warehouse Associate | June 2021 - Present

- Processed and organised shipments with accurate sorting, packing, and labelling to ensure efficient order fulfilment and workflow operations
- Maintained quality in a fast-paced environment through strong attention to detail and adaptability.
- Collaborated effectively with team members to support daily productivity targets and operational efficiency.